

Formulario

Fuerza eléctrica: $\vec{F} = q\vec{E}$

Campo eléctrico:
$$\vec{E} = \frac{1}{4\pi\epsilon_0} q \frac{\vec{r}}{|\vec{r}|^3}$$
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int dq' \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3}$$

Ley de Gauss: $\oint \vec{E} \cdot d\vec{S} = \frac{q_{\text{enc}}}{\epsilon_0}$

Potencial eléctrico:
$$\Delta V = - \int \vec{E} \cdot d\vec{l}$$
$$V = \frac{1}{4\pi\epsilon_0} \int \frac{dq'}{|\vec{r} - \vec{r}'|}$$
$$\vec{E} = -\vec{\nabla} V$$

Coordenadas polares: $0 \leq r < \infty, 0 \leq \theta \leq 2\pi, -\infty < z < \infty$
$$dl = r d\theta$$
$$dA = r dr d\theta$$
$$dV = r dr d\theta dz$$
$$\vec{\nabla} = \hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{z} \frac{\partial}{\partial z}$$

Coordenadas esféricas: $0 \leq r < \infty, 0 \leq \phi \leq 2\pi, 0 \leq \theta \leq \pi$
$$d\vec{l} = \hat{r} dr + \hat{\theta} r d\theta + \hat{\phi} r \sin \theta d\phi$$
$$dA = r^2 \sin \theta d\theta d\phi$$
$$dV = r^2 \sin \theta dr d\theta d\phi$$
$$\vec{\nabla} = \hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$$